

Customer Shopping Behavior Analysis

1. Project Overview

This project explores customer shopping behavior using a retail transactional dataset containing 3,900 customer records. The objective is to identify purchasing trends, customer demographics, subscription behavior, and product performance through Python, SQL, and Tableau. The dashboard is designed to help business stakeholders monitor customer behavior and support data-driven decision-making.

2. Tool Used

- Python
- Pandas
- NumPy
- PostgreSQL
- Tableau Public
- Jupyter Notebook
- Git

3. Dataset Summary

- Records: 3,900
- Features: 18
- Data Type: retail customer transaction dataset
- Key variables:
 - Customer information
 - Age
 - Gender
 - Location
 - Subscription Status
 - Shopping Behavior
 - Purchase Amount
 - Category
 - Shipping Type
 - Discount Applied
 - Review Rating
 - Purchase Frequency

4. Data Cleaning & Preparation (Python)

Data Cleaning

- Imported dataset using pandas

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

- Checked missing values
- Filled missing review ratings using median value by category
- Renamed columns using snake_case
- Removed redundant columns

Feature Engineering

- Created Age Group
- Created Purchase Frequency (Days)
- Converted categorical variables into standardized format

Data Validation

- Checked duplicated records
- Verified data consistency
- Prepared dataset for PostgreSQL

5. Data Analysis using SQL (Business Transactions)

Structured SQL analysis was performed in PostgreSQL to answer key business questions.

1. Revenue by Gender – Compared total revenue generated by male vs. female customers

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

Showing rows: 1 to 839					Page
	customer_id bigint			purchase_amount bigint	
1	2			64	
2	3			73	
3	4			90	
4	7			85	
5	9			97	
6	12			68	
7	13			72	
8	16			81	
9	20			90	

3. Top 5 Products by Rating – Found products with the highest average review ratings

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	avg numeric
1	Standard	58.4602446483180428
2	Express	60.4752321981424149

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status

	subscription_status text	total_customer bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

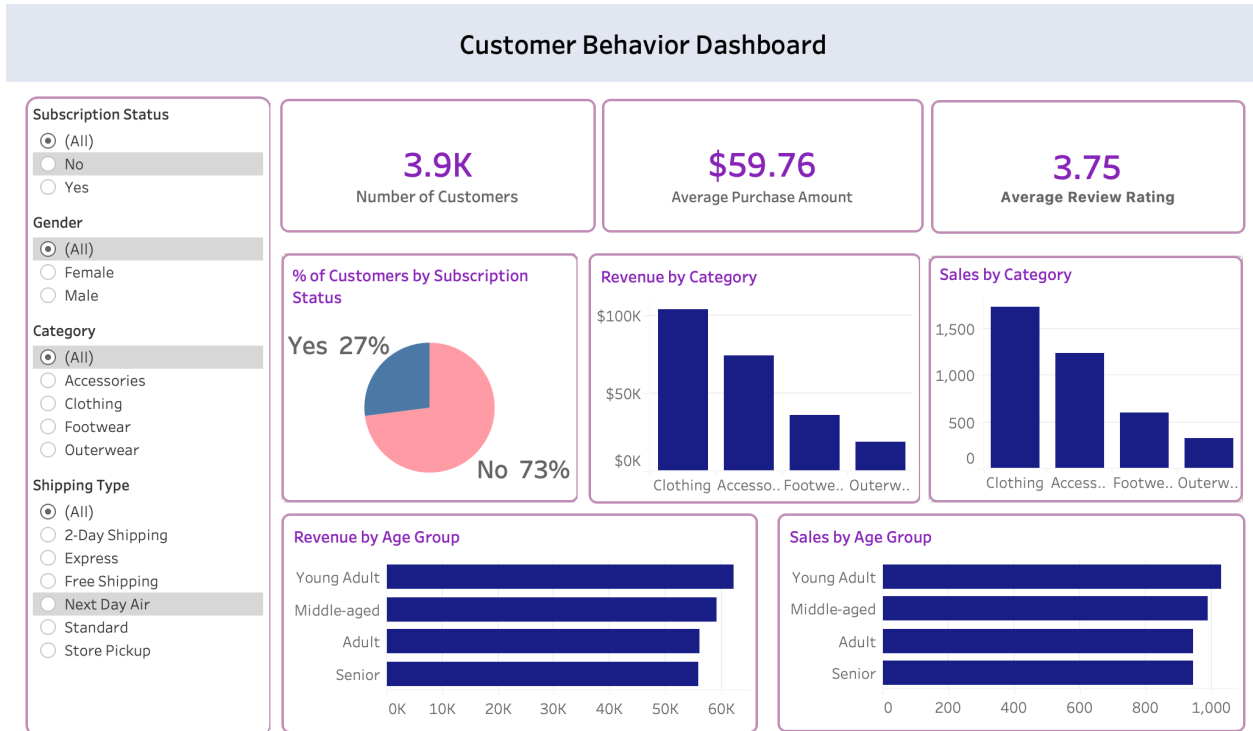
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

6. Dashboard in Tableau

An interactive Tableau dashboard was developed to visualize customer purchasing behavior.



7. Key Insights

- Clothing generated the highest revenue among all product categories.
- Young Adult customers contributed the highest total revenue.
- Approximately 73% of customers are non-subscribers.
- Average purchase amount remains consistent regardless of subscription status.

8. Business Recommendations

- Increase subscription conversion through exclusive member benefits.
- Allocate more marketing budget to Clothing products.
- Develop campaigns targeting Young Adult customers.
- Introduce loyalty rewards for repeat customers.
- Optimize shipping options based on customer preferences.

